Predicting Heart Disease with Support Vector Machines Using NHIS Health Behavior Data

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Research Question: Can demographic and health behavior variables classify whether a respondent has heart disease?

References:
Dataset Citation
Lynn A. Blewett, Julia A. Rivera Drew, Miriam L. King, Kari C.W. Williams, Daniel Backman, Annie Chen, and Stephanie Richards. IPUMS Health Surveys: National Health Interview Survey, Version 7.4 [dataset]. Minneapolis, MN: IPUMS, 2024. https://doi.org/10.18128/D070.V7.4

Software: Python, pandas, NumPy, matplotlib, scikit-learn

Theory: Technical Background of SVMs

Support Vector Machines (SVMs) are models that classify observations based on a set of predictors. The decision boundary the SVM uses can be linear, radial, or polynomial. In all three of these SVM kernel types, the C parameter controls the margin flexibility of the classification boundary. Gamma, used in radial and polynomial kernel SVMs, controls the influence of the kernel. Degree, only used in polynomial kernel SVMs, controls the model's complexity.

In this analysis, the three SVM models were tuned using cross validation using F1-score. The goal when tuning an SVM is to minimize classification error while maximizing the margin.

Methodology

Target: heart disease indicator

Predictors: sex, moderate physical activity, sleep, BMI, vegetable intake, and alcoholic beverage frequency

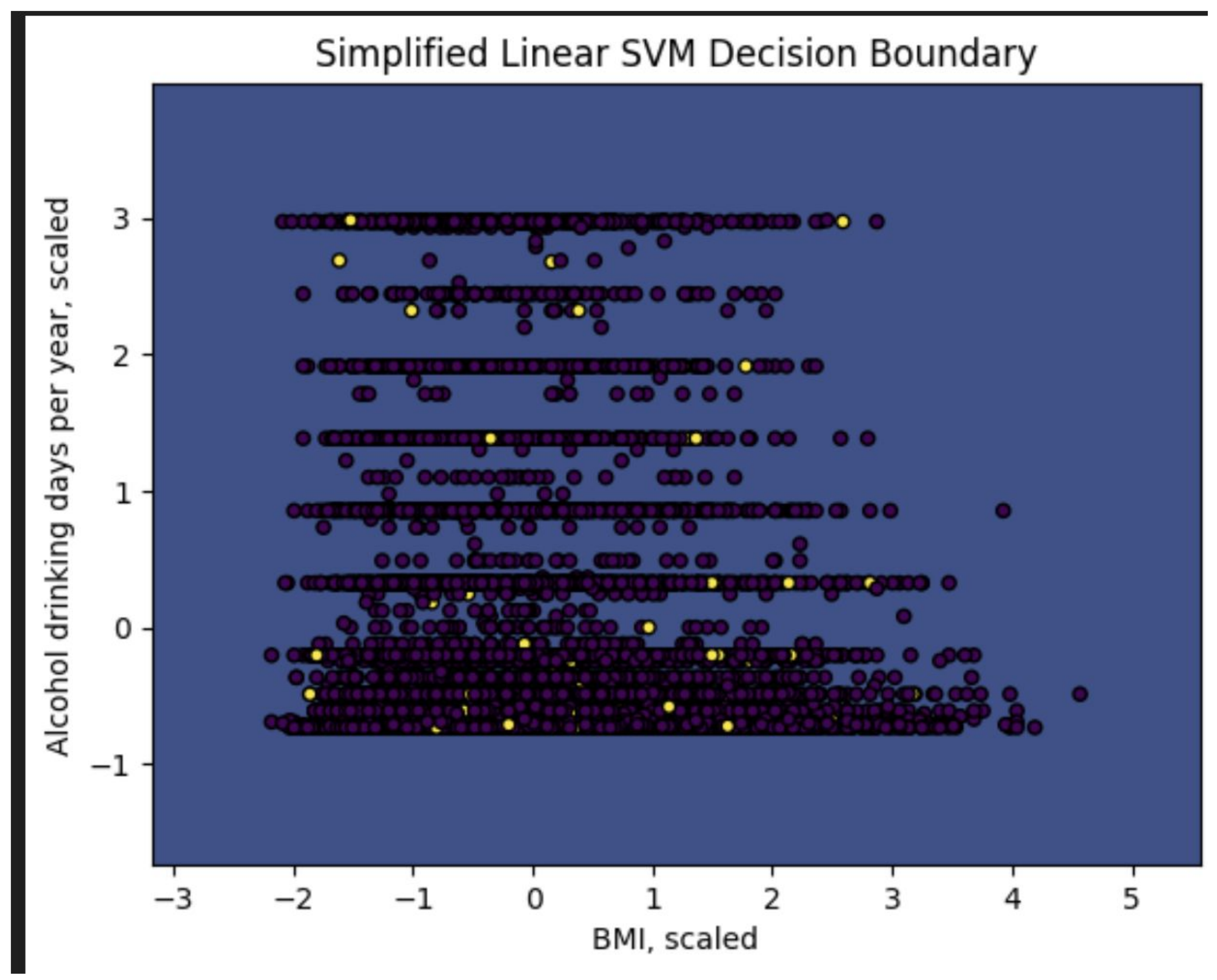
To clean the data, observations with invalid or missing codes, including missing target variable value, were removed.

The predictor 'SEX' was one hot encoded since it was a binary variable.

The data was split into training and testing sets.

The numeric predictors were scaled using StandardScaler.

Linear, radial, and polynomial SVMs were fit.



Class Balance:

No heart disease: 95.22%

Heart Disease: 4.78%

Class imbalance explains the misleading 95.22% accuracy.

	Model	Test Accuracy	Precision of Heart Disease Class	Recall of Heart Disease Class	F1 Score of Heart Disease Class
0	Linear SVM	0.9522	0.0	0.0	0.0
1	Radial SVM	0.9522	0.0	0.0	0.0
2	Polynomial SVM	0.9522	0.0	0.0	0.0

Simplified Linear SVM decision boundary

Discussion

Accuracy was misleading because the target class was severely imbalanced. The models mostly predicted the majority class. Therefore, the models did not identify many cases of heart disease. Evaluation metrics like recall and F1-score for the heart disease class were poor. BMI showed the clearest small difference between target classification groups. The moderate physical activity predictor showed a smaller difference. The selected predictors, overall, were not strong enough to separate the heart disease cases. Future work should consider using class weights, resampling, more/different predictors, and threshold based evaluation.

This analysis is a great reminder to not evaluate health classification models using just accuracy. In rare health conditions, recall and false negatives are metrics to evaluate.

Possible Improvements

Future modeling would have to address the class imbalance of the target variable (heart disease) more directly by using class weights, resampling, or tuning on another metric.

Trying additional predictors from the NHIS dataset may allow for more insight and clearer classifications.

Considering other models, like logistic regression or random forests, to compare to SVMs may also provide improvements.

Another possible method change would be to use adjusted thresholds to classify the observations. In practice, this would mean changing a default probability of greater than or equal to 50% being assigned the heart disease class to a user set threshold probability of, for example, greater than or equal to 35% probability. This could allow the model to identify more heart disease cases, increasing recall score for the heart disease class. However, lowering the threshold may increase false positives. This tradeoff would need to be balanced, which could be done by analyzing the evaluation metrics for various thresholds to find the optimal threshold. The other factor to consider with this suggestion is runtime, since calculating the probability of each observation is costly.

Conclusion

The selected health behavior and demographic predictors were not sufficient to classify heart disease well using SVMs. Although accuracy was high, the models failed to detect the minority heart disease class. Future work should address class imbalance and test additional predictors before using this type of model for health classification.